

Double Round Robin Tournament with Triples

Johannes Weiland

April 27, 2026

Description

The double round robin tournament with triples [1] is a variant of the classical double round robin format, where each team can have more than one match per round to reduce the number of necessary rounds in the tournament. Instead of single team matches, each round consists of minitournaments with three teams, called triples. Therein, the three teams meet at the venue of one of them and play a single round robin tournament, i.e., the triple (A,B,C) comprises the matches A – B, B – C, and A – C. After playing all matches of a triple, the away teams return home every time. The goal is to find a schedule minimizing the total travel distances of all teams over the season.

The problem can be described as follows:

Input: Given is a set of $n \in \mathbb{N}$ teams $T = \{t_1, \dots, t_n\}$ and the symmetric distance matrix $D \in \mathbb{R}^{n \times n}$.

Output: A schedule for a double round robin tournament on T , such that:

- All rounds should consist solely of triples. For $n = 3k + 2$, $k \in \mathbb{N}$, two additional single matches are necessary and scheduled in parallel to the triples.
- Each team can play at most one triple or single team match per round.
- The same three teams must not meet at the same home team twice.
- All teams should have the same number of home field advantages, if possible. Otherwise, the discrepancy between the home field advantages of any pair of teams should be minimal at the end of the season.

Example 1. For the simple case of three teams, we consider $T = \{1, 2, 3\}$ and

$$D = \begin{pmatrix} 0 & 145 & 24 \\ 145 & 0 & 143 \\ 24 & 143 & 0 \end{pmatrix}.$$

Writing each triple as a tuple with the home team in the first position, there are three possible triples $(1, 2, 3)$, $(2, 1, 3)$, and $(3, 1, 2)$ with associated total travel distances 338, 576, and 334, respectively. Since each triple can occur at most once, the optimal solution incorporates the triples $(1, 2, 3)$ and $(3, 1, 2)$ in arbitrary order. One optimal schedule is given in the following table.

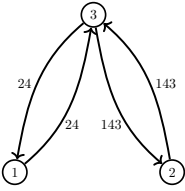
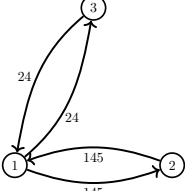
Round	Matches	Travel distances
1	3 – 1	
	1 – 2	
	3 – 2	
2	1 – 2	
	2 – 3	
	1 – 3	

Table 1: Optimal schedule for the example instance with three teams.

References

- [1] Johannes Schmidt, Armin Fügenschuh, and Moritz M Böschow. Sports league scheduling with minitournaments. *Cottbus Mathematical Preprints*, 33:1–17, 2025.